

Review

An Overview on patenting trends and technology commercialization practices in the university Technology Transfer Offices in USA and China

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ABSTRACT

The Technology Transfer Office is critical in translating university research into commercial value. This research paper analyzes the performances of university Technology Transfer Office models as well as their technology transfer mechanisms. The first step of this research project involves conducting patent information searches on databases to investigate academic patenting activities in the USA and China. Many Technology Transfer Offices, have similar intellectual property licensing practices and the involvement of industrial partners throughout the commercialization process. However, during the comparison it is found that the Technology Transfer Office working models varied in several ways. The main difference is that the pace at which patenting is accomplished in the USA and Chinese universities is far more sophisticated. Because of substantial resources and government support, Proof of Concept funding at various stages of technological maturity is steadily developing at universities in the USA and China. Furthermore, both USA and Chinese universities are very active in teaching and incentivizing their faculty and students about the importance of academic entrepreneurship and are well-versed with skilled technology transfer professionals who have all the necessary skills to transfer technology.

1. Introduction

In recent decades, universities have added a new dimension to their primary purposes of research and teaching with an additional purpose of commercialization of research-based technologies. Many research universities throughout the world have started establishing their Technology Transfer Offices (TTOs) also known as Knowledge Transfer Offices to manage and commercialize the university Intellectual Property (IP). TTOs act as a “facilitator” between innovators (university researchers, students, and faculty) and stakeholders such as corporations, entrepreneurs, and venture capitalists and help to commercialize the university-based innovations. On the plus side, licensing agreements and university-based startups can generate income for the university, job prospects for university researchers and graduate students, and local economic and technological spillovers by stimulating additional Research and Development investment and employment generation [1]. On the negative side, the expenses of TTOs may surpass the income gained through commercialization activities such as licensing, and there may be other unintended consequences, such as a possible shift away of faculty from academic teaching towards long-term research activities [2].

The industrial research stake in corporate research has declined by

8% between 1985 and 2015 according to a recent analysis as a result more industries and private research organizations rely on universities for research purposes [3]. This combines with the social effort to make the universities an important component of the economy by integrating the dimension of ‘transfer of knowledge’ to the aspects of the conventional teaching and research. Part of this difficulty stems from an ‘informative imbalance.’ On the one hand, the question of the quality of research is best informed by professors, and, on the other, the ability to use this scientific research, which are best informed for corporations. University TTOs are frequently the principal connections between universities and industries and so have the responsibility of exploring this imbalance of information and of matching academic research with industry. The TTOs are in the forefront in exploiting the potential involved in boosting universities participation in the marketing of their research. While many universities and industry have already begun to understand their mutual benefits based on successful partnerships for the transfer of knowledge among them, but Fig. 1 shows the most common dilemma which always remains between both the parties throughout the technology transfer process.

In addition, there is a persistent impression on the part of industry that the nature of academics and culture is not necessarily consistent with the requirements and goals of putting inventions on the market. Fig. 2 shows the challenges that may arise with people working from

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Table of abbreviations

TTOs	Technology Transfer Offices
PoC	Proof of Concept
US	United States
IP	Intellectual Property
EU	European Union
PCT	Patent Cooperation Treaty
US\$	United States Dollars
USA	United States of America
CN	China
WIPO	World Intellectual Property Organization

1.1. Role of TTOs in universities

The primary responsibilities of a university TTO are to encourage its inventors (faculty and students) to submit technological innovations to the TTO office, analyze those submitted inventions, and decide whether to apply for patent protection based on the evaluation of inventions considering the market value and maturity level of the technology. Most TTOs conduct the commercial and patentability evaluations of inventions based on a set of factors which are kept confidential. A typical TTO oversees maintaining the university's intellectual property portfolio, with inventors consisting of faculty and students. The TTO assists the commercialization of the university's IPs, either by licensing the IPs to third-party companies in exchange for royalties or by founding technology-based startups headed by professors or entrepreneurs. The TTO will assist the inventors throughout the technology transfer process.

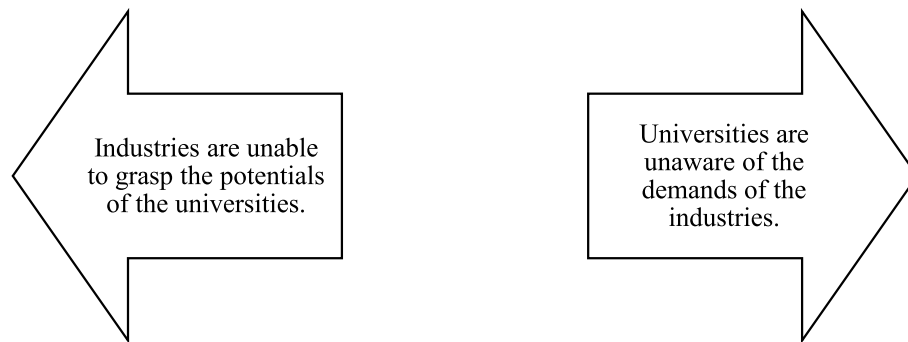


Fig. 1. Differences in understandings between industry and academia [4].

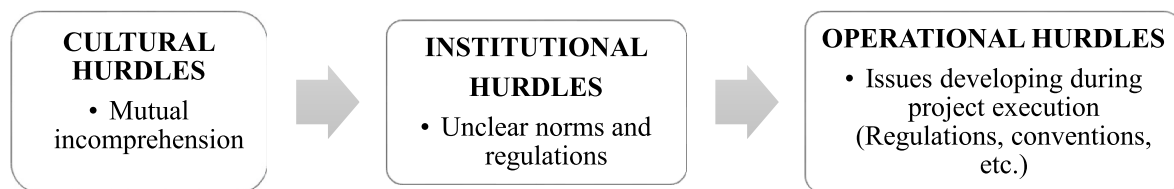


Fig. 2. Most common hurdles in the relationship between industry and academia [4].

Table 1

Various goals of industry and academia [5].

INDUSTRY OBJECTIVES	UNIVERSITY OBJECTIVES
New application for a real-time problem	An invention based on curiosity driven research
Added value to product/process	Knowledge advancement through investigation
Financial gains through the new application	New ways to pursue research
Applied Research	Basic research
Short-term development	long-term development
Research focuses on the product	Conduct research for the reasons such as • To be aware of research problem • What exactly it is about? • What is the reason behind this?
Confidentiality is maintained for a period	The public benefit is provided for free.
Intellectual property Protection/patents	Journal Publication/Conference proceedings
Commercial strategy	Freedom to study

academia and industries during the collaboration [4].

This is mostly due to the lack of information sharing between the two stakeholders. Table 1 provides a comparison of the differences in the goals and objectives of both the industries and universities.

It is important for a TTO to provide sufficient support to its inventors as it may boost the likelihood of them doing more significant research that can be monetized. A typical TTO in a university comprises of the units as shown in Fig. 3 [6].

1.1.1. IP management and technology licensing

A typical university IP management unit located anywhere in the globe is normally in charge of carrying out the activities outlined in Fig. 4:

- When an invention is disclosed or submitted to the TTO, the completeness of information essential for analyzing such inventions and perhaps deciding to file for patent protection is carried out in a systematic manner. Inventions are often dispersed among TTO personnel depending on research groups and TTO staff expertise. When a TTO receives an invention disclosure, the obvious first action is to gather and consolidate all essential information. This is usually arranged by the TTO and varies from university to university. This phase requires the completion of an Invention Disclosure Form [7].
- The technology transfer procedure starts when the faculty and students submit their inventions to the TTO by means of an official "Invention Disclosure Form" (IDF). The TTO will assign a docket

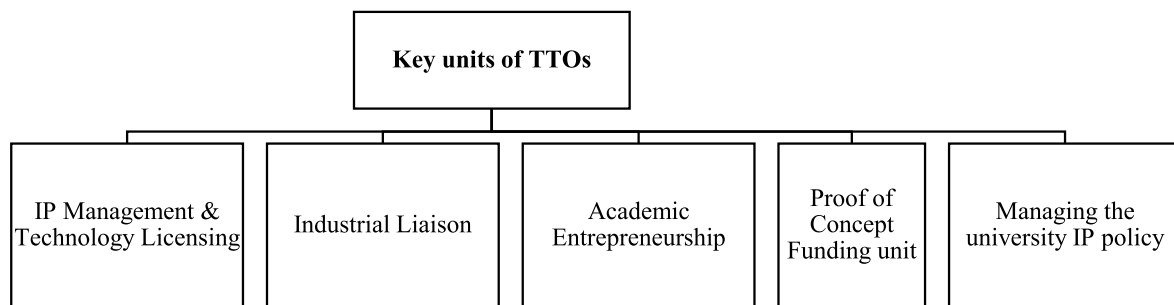


Fig. 3. Most common units in a typical matured university TTO [6].

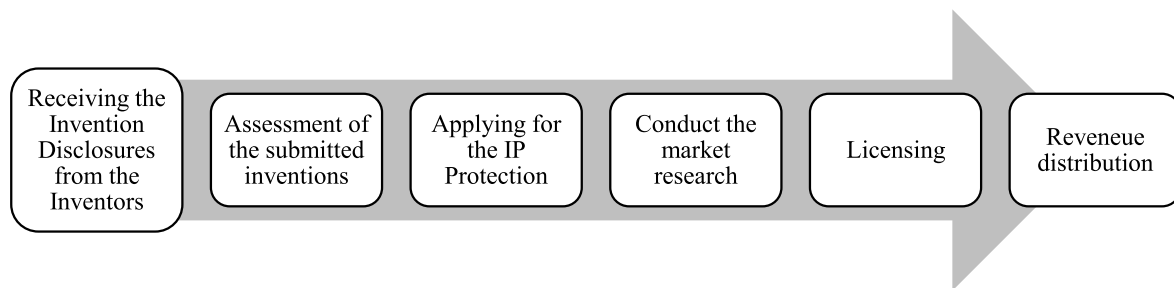


Fig. 4. Basic responsibilities of an IP management unit [7].

Invention Disclosure Form

- Submission at the TTO and should include all the information related to the invention, publications, inventor details and funding

Patentability Search (Inhouse/External)

- Identify the novelty and inventiveness in the invention and list out the key features

Drafting of the patent application

- Conduct the market search related to the invention and identify countries for protection
- Communication with industry professionals related to the potential of the technology and collaborations (NDAs)

Filing of the patent application at national or foreign Patent Office (PO)

- After the filing of the application at the PO, the applicant can approach the Venture Capitalists (VCs) for funding to upgrade the Technology Readiness Level (TRL)

Examination at the Patent Office (PO)

- Decision from PO (Acceptance/modification/Rejection)

Technology Commercialization

- Licensing the technology to companies/launching a technology based spin-off

Maintenance of Portfolio

- Payment of maintenance fees of the granted patents as well as monitoring the implementation of contracts. This also applies to maintaining know-how as trade secrets and safeguarding it through contracts.

Fig. 5. Typical route for IP protection and Licensing in a TTO [11].

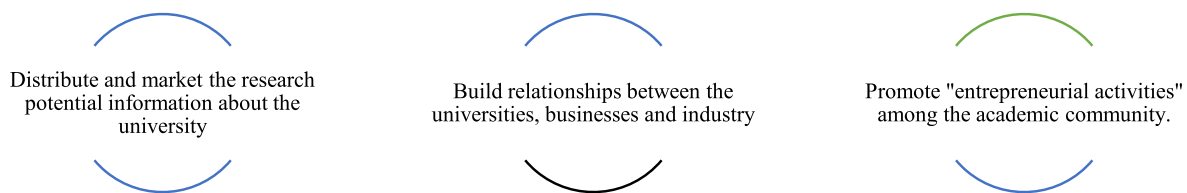


Fig. 6. Objectives of an Industrial Liaison office [12].

number to the IDF to aid in tracking the invention and use it in communications with the inventors and attorneys.

- The TTO will do a preliminary evaluation to identify the idea and will comprehend the research challenge addressed as well as how the conceived idea addresses such problem. Furthermore, they do a preliminary search on the invention to see whether the idea revealed in the invention meets the patentable criteria and has commercial value. The TTO staff will generate a brief report containing all these facts, as well as the patenting costs that may be required to apply for patent protection [8].
- It is critical to identify the correct inventors to reduce the risk of patent invalidation which usually occurs during the prosecution stage. The university TTO will ultimately play a key role in resolving ownership disputes among inventors affiliated with the university.
- The TTO typically evaluates the inventions which are disclosed in the office and further examines the market readiness and commercial potential of such inventions. The evaluation will let the TTO decide which commercialization option to take, whether go for licensing agreement or creating a spin-off company. Simultaneously, the TTO conducts a more extensive patent search to check whether the invention meets the patentability requirements or not [8].
- Following the evaluation step, the TTO will decide to apply for the patent protection of the invention since it is useful for forging licensing agreements and gives a very clear knowledge of the scope of the licensed technology. The patent protection starts with the filing of the patent application either in the domestic or foreign patent office [8].
- There are two commercialization pathways to get the technology to market, depending on the nature of the innovation, industry trends, finance availability, and interest from other companies [8]:
 - Collaborating with an established company by means of technology licensing agreement or other potential collaborations.
 - Launching Spin-off

With the collaboration situation, the TTO will be tasked with identifying the possible licensees/collaborators. Whereas the launching of a spin-off is regarded to be the superior approach, the TTO in this situation may look for funding to support the business [9].

- Licensing is an efficient and flexible way to build partnerships for universities with the industries thereby bringing the invention into the light of the market from the research labs. License is a “lease” arrangement between a licensee and a licensor based on authorization to use a patented invention in return to receive the compensation through the royalties. Licenses are applied to intangible assets such as the Intellectual Property (IP) protecting the technical innovation, creative designs, inventive processes, software’s, literary works that are patented, trademarked, copyrighted. Unlike tangible property, intellectual property (IP) can be licensed to numerous entities located in various jurisdictions at the same time. However, the lease/license is only valid for a limited time and is subject to the terms and circumstances of a mutually agreed-upon licensing agreement. These agreements also restrict how IP may be used to ensure that licensees only pay for the benefit of its usage and that licensors receive their “fair share” in exchange for each usage. This remuneration allows inventors to fund and conduct more research

Table 2

Key responsibilities of Industrial Liaison office [12].

INFORMATION DISSEMINATION	Responding to industry queries Creating and distributing flyers Creating possibilities for networking
PROMOTIONAL AND MARKETING ACTIVITIES	Coordinating laboratory visits Attending conferences and giving talks Attending specialized technology trade shows
ACTIVITIES INVOLVING PUBLIC AFFAIRS	Establishing contacts with professional organizations Publishing articles in magazines and newspapers Publicizing special events
ADVICE AND ASSISTANCE WITH RESEARCH CONTRACT NEGOTIATIONS	Providing intellectual property and patent advice to professors assisting with negotiations Developing a Technology Transfer Strategy
ACTIVE MANAGEMENT OF UNIVERSITY POTENTIAL ENHANCEMENT	Industrial partners searching Business Partner Search Financial partners searching (venture capital funds, business angels) Initiate spin-offs and support business strategies
UNIVERSITY RESEARCH PARK COORDINATION (IF AVAILABLE)	
UNIVERSITY INCUBATOR CENTER COORDINATION	
UNIVERSITY SEED CAPITAL FUND COORDINATION	

and development (R&D), hence sustaining the innovation cycle [10]. Fig. 5 shows all the tasks undertaken by the TTO for the IP management and licensing.

1.1.2. Industrial liaison office

Many TTOs are responsible for establishing and maintaining industry collaborations, which may be critical for collaborating and bringing inventions to market. Some universities have independent offices for industry and corporate relations, which often collaborate with the industries TTO. In this situation, TTOs frequently take advantage of the corporate relations office’s ties, concentrating on the technology transfer process itself. The key mission of the Industrial Liaison Office can be classified as shown in Fig. 6:

The combination of university expertise with industrial financial assistance has resulted in several breakthrough innovations as well as the establishment of new companies, employment, and wealth. These results, however, are not always straightforward to achieve. Successful collaboration between industry and academia necessitates a certain type of synergy. To reach a successful collaboration agreement, both stakeholders must understand each other’s interests and objectives, as well as their complementing abilities. Table 2 highlights some key duties performed by the university industrial liaison office in an enabling innovation ecosystem in universities [12]:

Interaction between industry and universities is the first step in technology transfer. The relationship between industry representatives and university faculty, follows a regular process as shown in Fig. 7.

Fig. 8 shows the common practices used by industrial liaison office

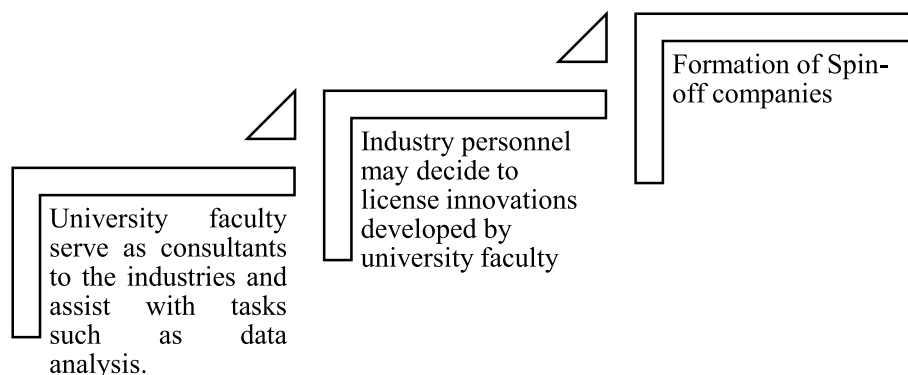


Fig. 7. Typical collaboration route between Industry and Academia [13].

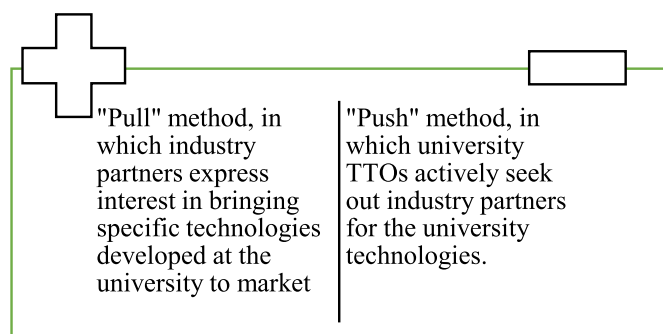


Fig. 8. Engagement strategies used by universities TTOs with industry partners [13].

during the technology transfer between university and industry.

1.1.3. Academic entrepreneurship

Academic entrepreneurship aims to foster an entrepreneurial attitude, skills, and characteristics in students, professors, and entrepreneurs so that they can recognize business possibilities out of their inventions. Business incubators, which are initiatives meant to encourage the successful growth of entrepreneurial/start-up enterprises, are commonly seen in academic entrepreneurship. This is accomplished through a variety of business support services and resources developed and organized by the unit and made available both within the incubator and through its network of connections. The academic entrepreneurship section oversees instilling an entrepreneurial mindset in students, faculty and researchers through teaching, training, mentoring and

research, as well as providing technical and financial assistance for the establishment and growth of high-potential spin-off businesses [14]. Academic entrepreneurship strives to educate university innovators (faculty, students, and researchers) on the issue of forming a firm to market the outcomes of their research. Today's academic entrepreneurship key responsibilities can be summarized as shown in Fig. 9 [15]:

1.1.4. Proof of concept funding

Proof of Concept (PoC), also known as proof of principle is a way to establish the viability of a technical invention by using a specific approach or idea. It is in principle a demonstration to check that certain concepts or theories possess practical potentials and can be used in the industries. The availability of its own foundation for supporting innovative projects is a key feature of the university PoC funding unit. One of the primary purposes of the TTO is to attract funding for the development of prospective high technology projects. As a result, universities scope initiatives before presenting them to investors. In a circumstance when the investment circle is large, it is necessary to address the individual needs of everyone. This creates difficulties at the application stage due to the necessity to comply with conflicting formal criteria. Investors, for their part, can only work with the projects that are offered to them and are frequently dissatisfied with the quality of a project's preparation. Creating the functional of project planning and finance inside the same structure is one technique to increase the efficiency of working with innovative projects [17]. The PoC financing model is widely employed in universities in the United States. In 2001, the William J. von Liebig Foundation granted the William J. von Liebig Center a donation of \$10 million to the University of California San Diego (UCSD) Jacobs School of Engineering. The stated mission of the von Liebig Center is to expedite the commercialization of UCSD technologies, foster and encourage the exchange of ideas between academia and business,

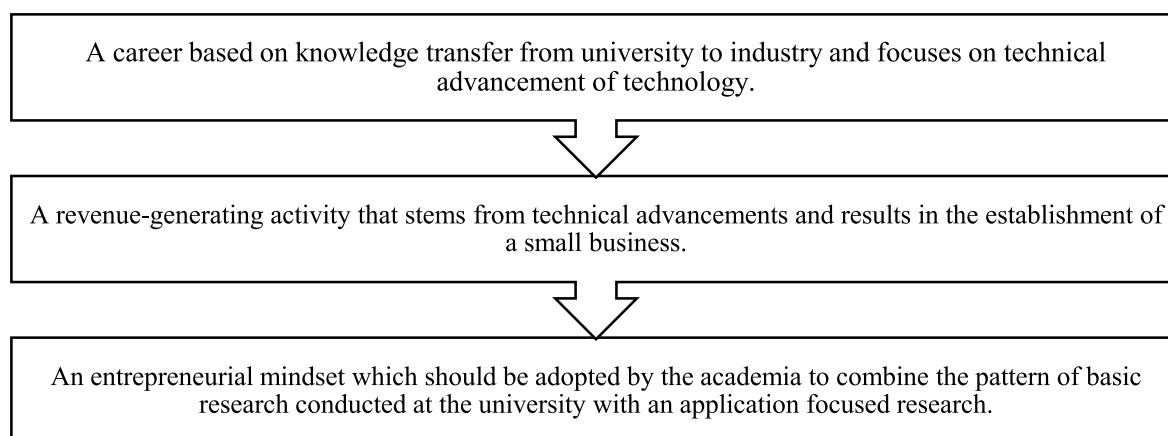


Fig. 9. Key responsibilities of an academic entrepreneurship center [16].

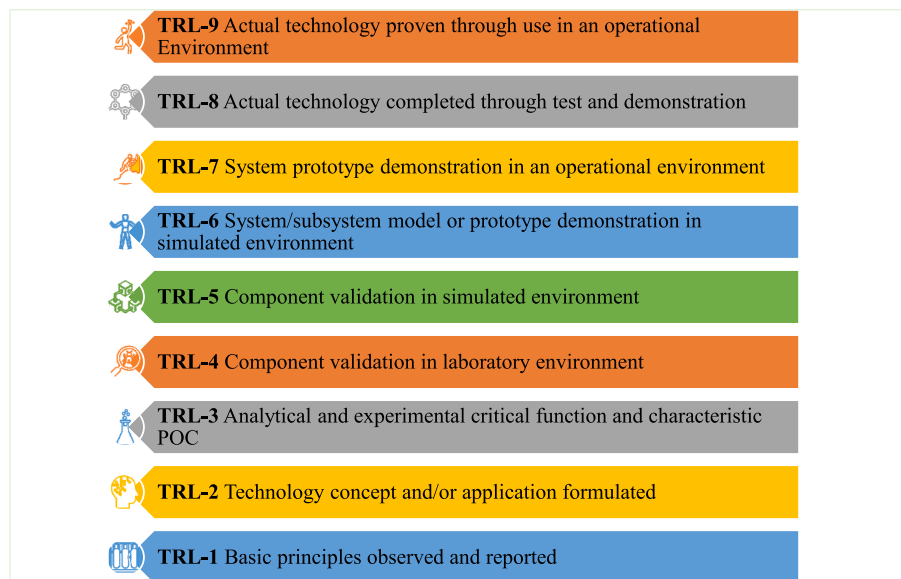


Fig. 10. TRL levels with description [19].

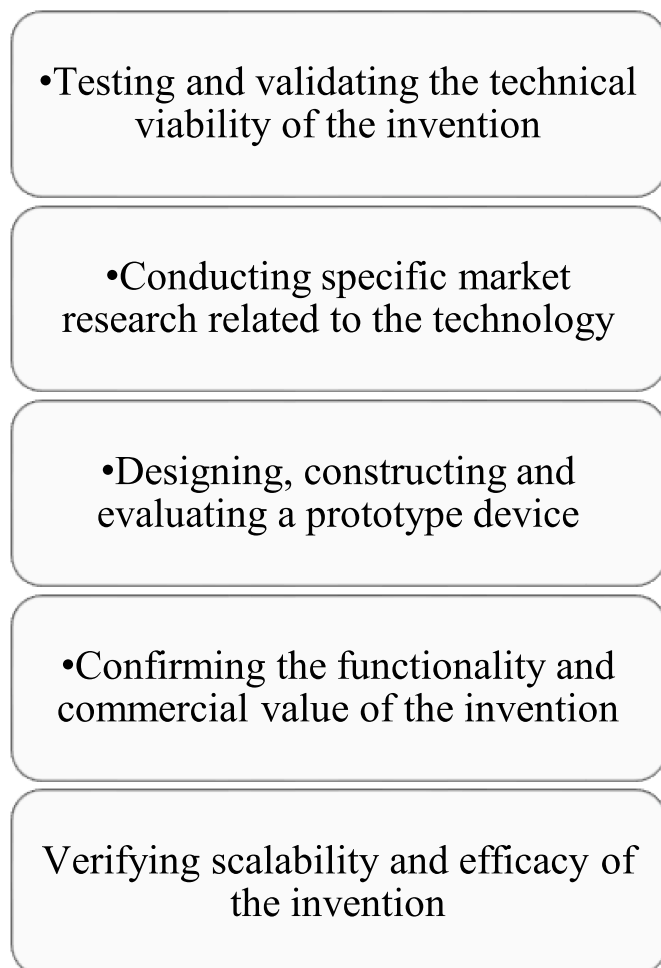


Fig. 11. Job responsibilities of PoC employees [20].

and prepare engineering students for the entrepreneurial environment. In 2002, Jaishree and Gururaj Deshpande donated US\$17.5 Million to the MIT School of Engineering to establish the Deshpande Center. “The Center was established with the goal of increasing the market impact of

innovative technologies developed at MIT [18].

The university’s PoC unit assesses the inventors’ innovative proposals based on its Technology Readiness Levels (TRL) which was developed by NASA. TRL evaluation is a way of determining the technical maturity of critical technology elements of a program. They are decided as part of a technological readiness assessment, which looks at program concepts, technological needs, and demonstrated technology capabilities. TRL is a scale of one to nine, with nine being the most developed technology. TRLs provide for consistent and comprehensive assessments of technological maturity across a variety of technologies. When analyzing program risk, the PoC staff who are the decision-makers will take the suggested TRLs into account. Evaluations can aid in assessing the advancement of technological development, informing program strategies, and identifying possible issues for decision-makers. Conducting TRL assessments at the early stages of development helps uncover possible issues before they become problems in the later and more expensive stages of system development. TRL can also enhance communication throughout developments and at important points of decision by offering a common language to communicate the readiness of technology and related technical hazards between technical developers (inventors), program managers (TTOs) and licensees’ team officials. Finally, the evaluation results can feed the planning activities such as estimates of costs and schedules, risk assessments and technological maturity plans. Fig. 10 shows the TRL levels of technologies with the explanation [19].

Usually, the basic research conducted at the universities falls under TRL-1 and TRL-2 and requires PoC funding for technology advancement and demonstration. Fig. 11 shows some common activities which are performed by the PoC staff during this process.

1.1.5. University IP policy

The possible ownership of IP and the prospect of moving it into a commercial stage is an increasing motivation for universities to pursue the technology transfer process. A university IP policy is essential to guarantee the effective transfer of technology between academics and commercial partners in the industry. Without a comprehensive, legally enforceable policy governing IP ownership, all parties participating in the technology transfer process would lack legal clarity when making IP decisions. Furthermore, having an IP policy simplifies compliance with key rules and regulations, as well as balancing conflicts of interest among the university, industry, and society. The TTO is responsible for managing the university IP policy. After the Bayh-Dole Act was

implemented in the United States, universities began to develop and constantly alter their intellectual property policies [21]. Following the act's implementation in 1981, universities replaced innovation equity with ownership rights. Some universities further broadened the concept of "inventions" to include "inventions that are not patentable," but others included inventions, copyrights, and data under the umbrella term "intellectual property." It is typical for TTOs at North American universities to establish an IP policy that extensively covers the legalities linked to participation agreements, disclosure and reviews, inventions (including student innovations), and return of innovations. In terms of technology transfer, the policy governs the licensing requirements, the distribution of future revenue from inventions, and the allocation of the equity pool. Finally, to avoid conflicts of interest, they create guidelines controlling the inventors' research activities as well as the usage of tangible research property within the university [22].

2. Assessment of academic patenting in different regions

In this research study, the author has investigated the academic patenting culture and the TTO model of top university located in the United States and China. The objective of the research was to identify the top universities in each region that are active in registering patents in recent years and to investigate the TTO model of these universities. To get the statistics, the author conducted patent information searches utilizing free patent databases such as Patentscope from WIPO and Espacenet from the EPO. These databases contain all records of granted and published patents filed at the European Patent Office as well as most national patent offices across the world. It includes information such as the filing and grant dates, the applicant's name, legal events, citations, and classification data. The study's conclusion was derived utilizing data from both the patent databases. Additionally, 3D landscapes diagrams were created using the patent database "Patsnap", which is based on patent analysis techniques. The database examines numerous characteristics inside the specified patents to study, and then compares the resemblance of each patent. Once the algorithm recognizes the resemblance, it creates a map of the technical spaces covered by the selected patents. While peaks and troughs indicate regions of activity, closeness indicates the degree of similarity between patent applications. The parameters are compared based on their IPC and CPC similarity, as well as keyword resemblance. Patents that are similar are placed closer together, whereas patents that are distinct are placed further away. PatSnap will generate a mountain in that space and label it with the most common keywords identified in that space if many comparable patents form a cluster. As a result, the peaks, where patents will be grouped, reflect places of high activity, while the low-lying parts reflect areas of lesser patent activity [23].

2.1. Universities with highest number of US patents grants

The universities have redesigned their technology transfer capabilities to capitalize on patent system reforms such as the legal right granted to universities in USA to own the assignment to the inventions made with federal research grants through the Bayh-Dole Act of 1980. The growth in patenting and licensing operations in USA universities is commonly considered because of legislation reforms such as the Bayh-Dole Act, as well as the advancements in the various breakthrough technologies which shifted the university research aim towards an industrial commercialization perspective. Nowadays, the focus of many universities has mostly been on generating the revenues through the patent licensing process, although only a few universities have been able to make considerable revenue from the technology transfer practices. In truth, most university TTOs do not earn enough revenue to pay their operational costs whereas the research financing organizations focuses explicitly on how much intellectual value is translated into economic worth and how the university interfaces with Industry. Before 1980, less than 5000 patent families were granted to universities based in USA

[24].

The Bayh-Dole Act, also known as the Patent and Trademark Law Amendments Act, is a part of USA legislation that deals with inventions resulting from federally supported research. Prior to the passing of Bayh-Dole, the USA government had amassed 28,000 patents, but only about 5% of all those patents had been commercially licensed. The Bayh-Dole Act was enacted in part to address the low usage of government patents. Before the enactment of this Act, the government held ownership to innovations created with government funding whether the research was conducted in federal laboratories, universities, or by independent enterprises. Licenses to use government patents were then negotiated with enterprises on a non-exclusive basis or in rare cases, for exclusive use by one company. However, it was frequently believed that without rights to an invention and the protection a corporation would not devote the significant, time and money required to develop a product or method for the market. The Bayh-Dole Act was intended to provide a single, consistent national strategy to reduce bureaucracy and encourage private business to use government-financed innovations by committing the risk capital required to develop such inventions to the point of commercial use. The patent system intended to be used to improve collaboration between academia and the business sector to ensure that ideas are brought to market. The Act also calls for expanded engagement of small businesses in the R&D industry, based on the idea that smaller businesses are more inventive than larger corporations [25].

The Bayh-Dole Act's revisions to patent regulations aimed to facilitate joint endeavors within and among academics, business, and government. Universities provided 14% of R&D in the USA with most of this being directed towards the foundational research which is essential to technological advancement. The work is completed as part of the educational process and serves to teach scientists, engineers, and managers who will later work in the commercial sector. Universities, on the other hand, do not typically have the industrial capabilities required to turn research findings into commercial goods on the other hand industry is responsible for such operations. As a result, the Bayh-Dole Act places a premium on encouraging collaboration between academics and the business community. The Act is designed to encourage the two sectors to collaborate to develop innovative goods, processes, and services for the marketplace by giving universities with IP ownership with which to explore and organize joint ventures. This type of collaborative work provides for shared expenses, liabilities, resources, and skills. The Act was sponsored by two senators, Birch Bayh of Indiana, and Bob Dole of Kansas and was passed in 1980 and enacted at Government Patent Policy Act of 1980 and in Chapter 18 patent rights in inventions made with federal assistance 35 U.S.C. 200–212. The Act is enforced by rights to inventions made by nonprofit organizations and small business firms under government grants, contracts, and cooperative agreements (37 CFR part 401) and licensing of government-owned inventions (37 CFR 404). Prior to the Bayh-Dole Act, the federal procurement regulation required the use of a patent rights clause as per which the inventors were obligated to assign innovations developed under contract to the federal government unless the financing agency determined that the national interest would be better served by allowing the inventor to retain principle or exclusive rights. The National Institutes of Health, the National Science Foundation, and the Department of Commerce had all developed mechanisms that permitted non-profit organizations such as universities to retain ownership of innovations after receiving notification and without having to seek an official judgement [26].

In an academic context, ownership of inventions is supposed to motivate the university to license the technology to businesses for commercialization in exchange for royalty payments. The Bayh-Dole Act has been seen as exceptionally successful in achieving its goals. However, while the act establishes a broad framework for expanding the use of the findings of federally supported research and development, concerns have been expressed about the appropriateness of present arrangements. Most people think that increased collaboration between business, government, and academia may boost financing sources (both

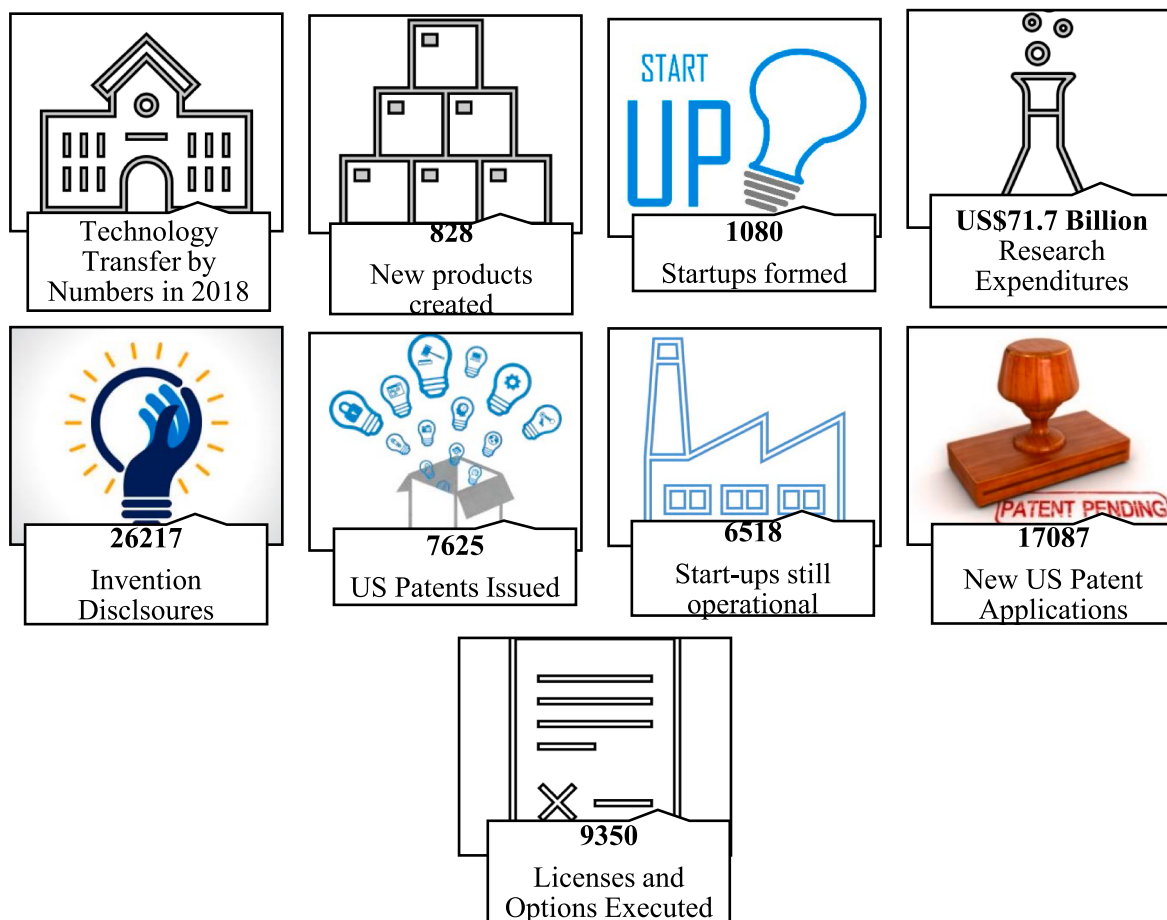


Fig. 12. Autm statistics in 2018 [29].

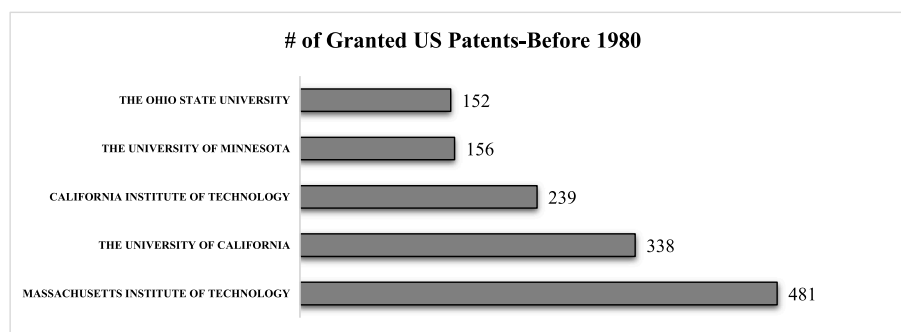


Fig. 13. Universities with the greatest number of granted patents before the passing of Bayh-Dole act.

private and public), boost knowledge transfer, inspire greater innovation (beyond invention), lead to new products and processes, and extend markets. Others, however, argue that collaboration may raise the likelihood of conflicts of interest, research misdirection, less openness in sharing scientific findings, and a larger emphasis on applied rather than basic research. Concerns have also been raised, notably in the pharmaceutical and biotechnology industries, that the government and the public are not obtaining benefits associated with the federal commitment to the original research and development [27].

In the USA, almost US\$71 Billion was spent on federally sponsored research at universities in 2018 and over US\$2.94 Billion in license revenue was made directly through the process of bringing academic ideas to market, often known as Technology Transfer. The USA government have invested more than US\$100 Billion in research funding

each year, with a total investment of more than a trillion dollars during the previous 15 years. Fig. 12 shows some key indicators published by Association of University Technology Managers (AUTM) in 2018, demonstrating the impact of university technology transfer in the USA [28].

According to patent information gathered from databases, prior to the passage of the Bayh-Dole Act in 1980, less than 5000 standalone patents included universities as the owner/applicant in the granted patents. Fig. 13 shows the top universities with the highest number of granted US patents with the issued date before 1980.

Fig. 14 shows the top IPC classification for the US patents with the assignees as universities before 1980.

Following the passing of the Bayh-Dole Act in 1980, there was a significant rise in the number of patents granted to US universities. From

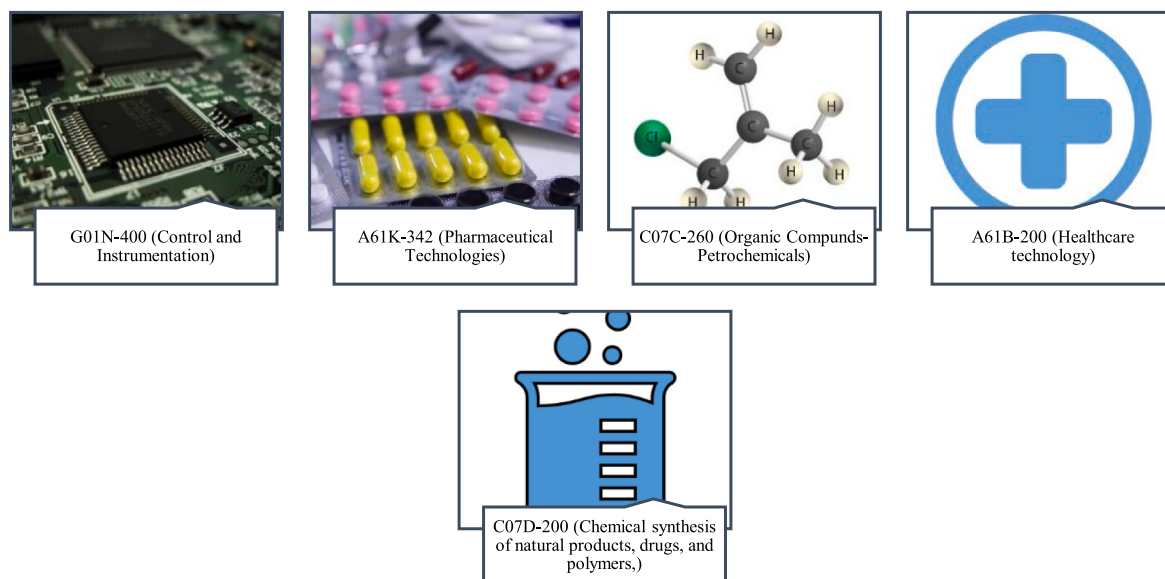


Fig. 14. Top IPC classification before 1980.

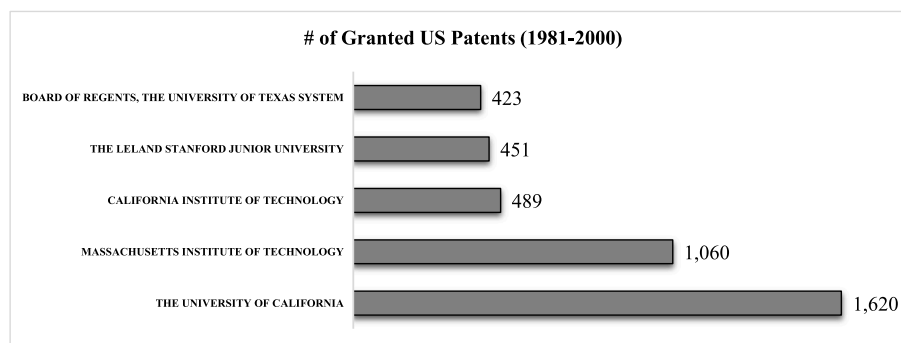


Fig. 15. Universities with the greatest number of granted patents between 1981 and 2000.

A61K-5126 (Pharmaceutical Technologies)
C12N-3302 (Genetic Engineering)
G01N-3068 (Control and Instrumentation)
C07K-2944 Peptides (Muscle and bone enhancements in humans and animals)
C12Q-1544 (Biotechnology)

Fig. 16. Top IPC Classification for patents owned by universities between 1981 and 2000.

1981 to 2000, Fig. 15 depicts the universities with the highest number of patent grants. As per statistics extracted from the databases, around 23,000 standalone US patents were granted to US universities during this period.

Fig. 16 shows the top IPC classification for the US patents with the assignees as universities during 1981–2000. During this time, the IPC-A61K, which relates to pharmaceutical technology, obtained the most patent grants for US universities followed by C12 N which is associated with genetic engineering technologies.

More than 77,000 patents were awarded to universities in between 2001 and 2020. Fig. 17 shows the top universities with the highest number of patents grants. During this time span, the University of California received more US patents than any other university.

Fig. 18 illustrates the top technology sectors in which universities sought patent protection. During this time, the IPC-A61K, which is closely tied to pharmaceutical technology, obtained the most patent grants for US institutions.

2.1.1. TOP University technology transfer office in USA-BASED on patents grants-university of California

The TTO in University of California (UC) comes under the supervision of "Research & Innovation". The TTO manages the ten campuses of University of California. According to the Bay Area Council Economic Study, enterprises developed and linked with UC contribute US\$20.1 billion in overall economic activity and linked with over 38,000 employments to the US economy. In recent years, the TTO has experienced a surge of innovation and entrepreneurial activity because of the huge financing it received through many governments and private sources, especially the Assembly Bill (AB) 2664 in 2016. The AB bill awarded each campus of UC with US\$ 2.2 Million to strengthen the university entrepreneurial facilities and introduce new specialized education programs. Through the AB funding alone the campus has created more than 60 incubators, accelerators, entrepreneur training programs, academies, and pitch contests that help startups, enterprises, and social entrepreneurs. The following services were made available to UC entrepreneurs through the AB financing bill [30]:

- More funding for proof of concept

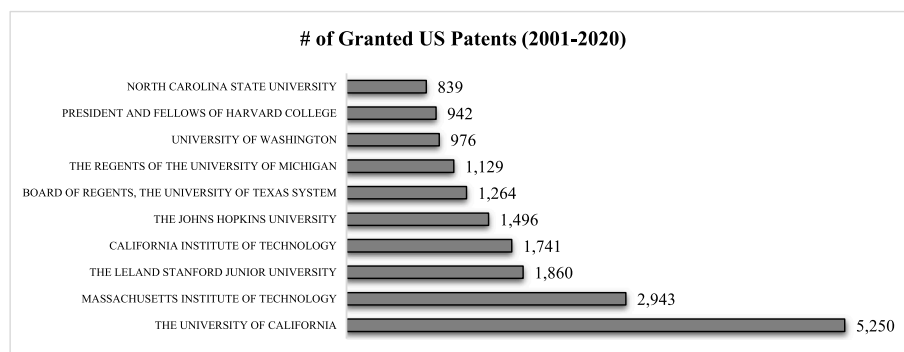


Fig. 17. Universities with the greatest number of granted patents between 2001 and 2020.

A61K-18,119 (Pharmaceutical Technologies)

G01N-11054 (Control and Instrumentation)

C12N-10025 (Genetic Engineering)

C07K-8107 (Muscle and bone enhancements in humans and animals)

C12Q-5959 (Biotechnology)

Fig. 18. Top IPC Classification for patents owned by universities between 2001 and 2020.

- A growth in the number of accelerators and incubators for university entrepreneurs where the teams can create, test, and launch their prototypes or companies.
- Additional mentoring and training programs to faculty and students thereby assist them in their business endeavors.

Fig. 19 shows the primary goals of the TTO as described on their university website.

The following are the different sections which are available in the UC-TTO and are assigned with various responsibilities:

2.1.2. Office of innovation & entrepreneurship

- This division strives to improve the university community's innovation skills and to develop the entrepreneurial mindset. It sponsors academic programs and forms collaborations with the government, business, and charitable sectors to achieve benefit to society and economic progress.
- The people who work in this unit aspire to generate possibilities and promote knowledge via technologies that benefit society.
- Based on UC Technology Commercialization Report, this unit have successfully managed to launch 1218 startups since 1980–2018 [30].

- The staff in this department form relationships beyond the research laboratories and classroom to make practical outcomes available to the public. They participate in the following activities [31]:

Identify opportunities that can be transformed into valuable goods and services.

Carry out assessments, data collection, proof-of-concept studies, or prototypes to establish commercial viability.

Conduct market research and build an entrepreneurial team to get funding.

Assist with company incubation at the early phases of conception and growth.

2.1.3. Knowledge transfer office

- Based on the information published on the university website, the university achieves technology transfer by educating students, publishing research results in journals and conferences, and ensuring that innovations are transformed into viable products in the commercial marketplace [32].
- The university has a well-established TTO which manages the university IP portfolio and looks after the IP management and commercialization activities, especially related to licensing of technologies [33].
- The TTO website also shows the technologies which are available for licensing with the third-party companies and for the students interested in launching technology-based startups. They have designed a smart interface on their website to select the technologies based on categories, many subsections for the main technology category, inventor name, department name and search by keywords. Each invention under the technology category is briefly described with its applications and the concerned TTO staff [33].
- Currently, University of California operates on two IP policies with regards to terms related to revenue sharing with the inventors. Based on certain conditions as described on the university website, the university employees will have the option of selecting their inventions governed under the Old Policy or the new Policy, but it must be defined clearly defined when submitting the Invention Disclosure Form in the TTO. The complete text of each policy may be seen on the university's website under Research Policy Analysis and Coordination.

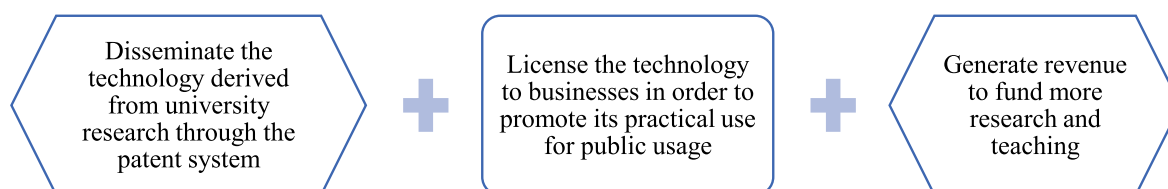


Fig. 19. Primary goals of the TTO [30].

Total inventions submitted in the TTO: 1735
Total applications filed: 1908
Total US patent grants: 615
Number of new startups: 93
Total technology commercialization income: US\$200.6 Million

Fig. 20. UC-TTO statistics for the fiscal year 2018 [34].

The former policy distributed royalties to inventors at a rate of 50% of net income, but the current policy distributes royalties at a rate of 35% of net income to university inventors. The TTO clearly explains the royalty sharing calculations on their website with some examples [34].

The TTO employs a total of twenty-two people, who perform various tasks such as Prosecution of patents, financial management, business systems, projects and Key performance indicators, Technology marketing, and employee tools and resources. Fig. 20 illustrates the UC-TTO's commercialization activities and revenue generation for the financial year 2018.

Patent classification codes are a sort of patent categorization that is used to uniformly classify the technology disclosed in a patent. IPC codes are used in over 100 countries and are managed by the World Intellectual Property Organization (WIPO). Cooperative Classification (CPC) is also used for technology classification. The CPC is managed jointly by the European Patent Office (EPO) and the United States Patent and Trademark Office (USPTO). CPC codes are now being implemented in 39 different countries. The information about the IPC and CPC codes is normally included in published patent documents [35]. Fig. 21 represents the top IPC and CPC subclasses codes for all the patent grants for University of California in 2020. The IPC and CPC codes with the highest numbers correspond to technological areas having a high level of patent activity.

The main technological categories for which the University of California has patents are depicted in Fig. 22, and the information was obtained from the university's website [36]. Fig. 22 shows that the

university has a diverse portfolio of technologies in areas such as medical, biotechnology, research tools, materials, and chemicals, and so on. The top IPCs and CPCs indicated in Fig. 21 correspond to the major technologies offered for licensing by the university on their TTO website. These analytics are crucial for analyzing and comprehending an organization's technological advancements.

2.2. UNIVERSITY technology transfer offices in China

The official Intellectual Property (IP) law was first adopted in the People's Republic of China in the 1980s, and it has since been progressively developed. IP was a new legal and social term for the People's Republic of China at the time. However, because of recent attention, social awareness of intellectual property rights in China has increased. Following a set of ministerial and commission regulations on technology transfer, universities today generally own the IP arising from the government-funded research and have the authority to use the IP for the revenue generation. The Ministry of Science and Technology and the Ministry of Finance jointly released a landmark measure for Intellectual Property created under the Government Funding in 2002, which is also referred to as the "Chinese Bayh-Dole Act." Fig. 23 indicates some significant powers which were granted to the Chinese universities related to the IP conceived through the government funding [37]:

Fig. 24 shows the important details of the Article 6 in the patent law of the China as it defines the basis of the ownership of patents conceived by individuals/faculty while working in a company or university [38].

Since then, the number of patent applications submitted by Chinese universities have increased drastically. Both the filed patent applications and granted patents to the universities increased significantly after the introduction of this improved IP rights legislation by the ministerial and commission regulations [39]. Fig. 25 illustrates the data extracted from the Patsnap database related to the patent filings by the Chinese universities in China National Intellectual Property Administration when compared to the total filed applications over the years.

Additionally, during the period from 2016 to 2020, the Chinese universities have received ~568,825 patent grants of the total grants issued by the China National Intellectual Property Administration which is the official patent office in China (Total Grants: ~2,256,382). Fig. 26 shows the percentage distribution of patent grants by the Chinese patent

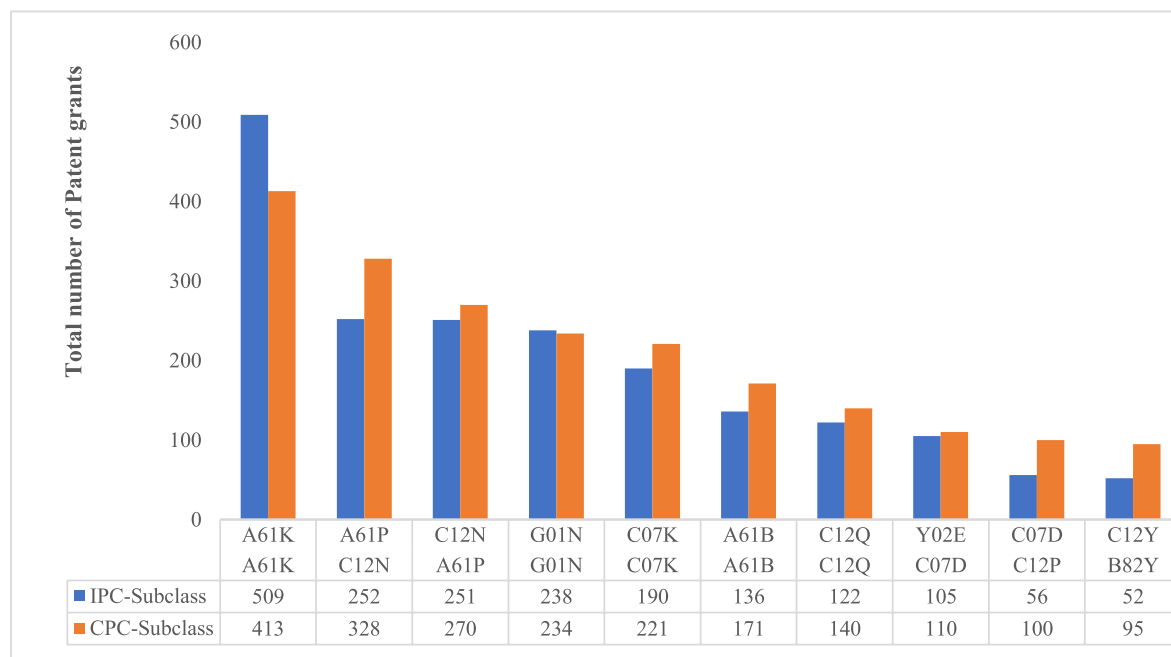


Fig. 21. Top IPC and CPC Subclasses with codes for all the patent grants for UC in 2020.

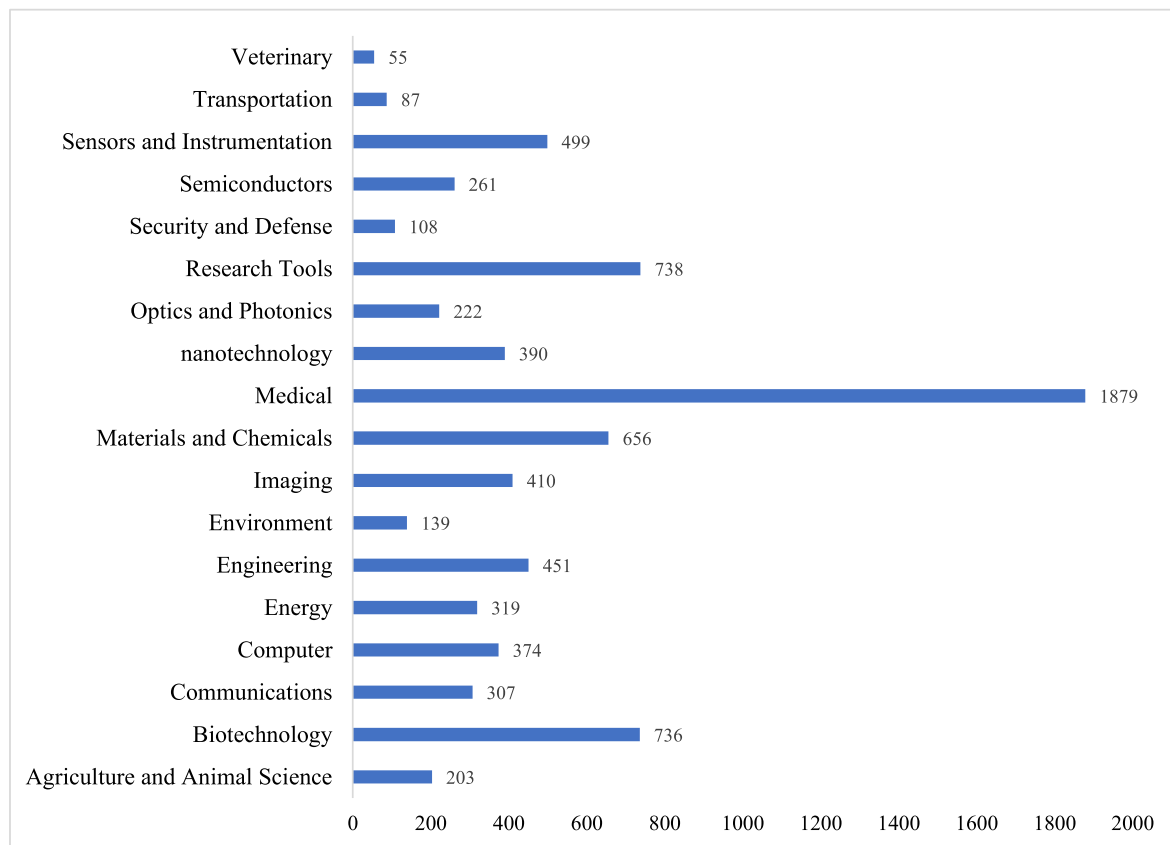


Fig. 22. Technologies owned by UC and available for licensing [36].

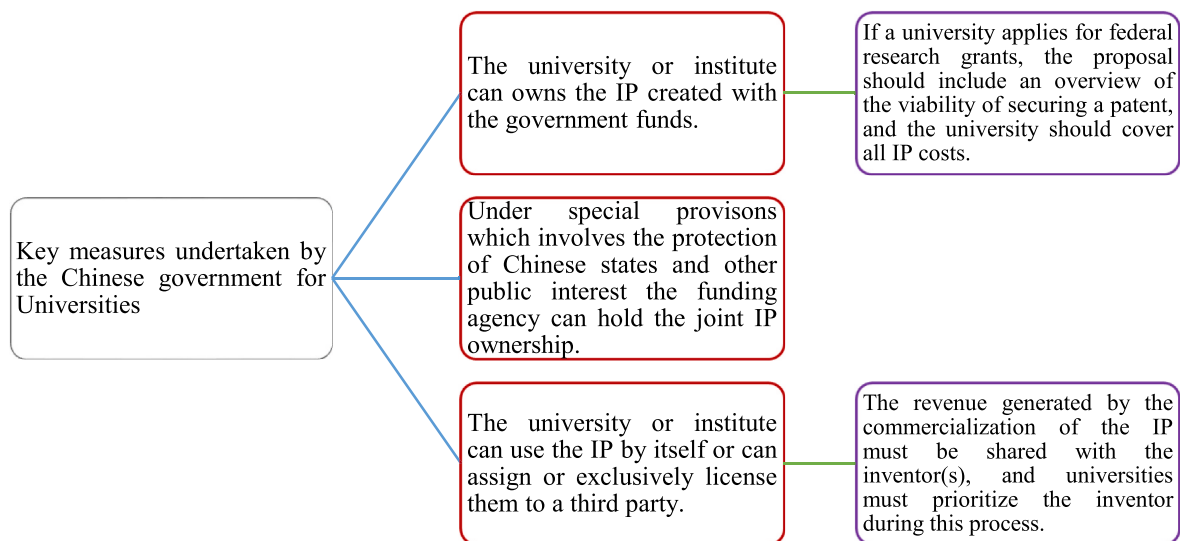


Fig. 23. Key powers granted to universities from the legislation [37].

office for various entities:

This rise in numbers is primarily attributed to an increasing understanding and appreciation of the importance of IP in Chinese universities, which has been followed by a growing acceptance of the concept of IP in Chinese culture. However, those statistics may represent a pattern in which researchers and organizations use patents to enhance their reputations rather than to pass or commercialize technologies. Most universities still lack institutional IP policy and autonomous TTOs. Nonetheless, several world-class Chinese universities have mastered the

art of IP management and commercialization. The top universities are both an anomaly and a model for other Chinese universities, having effectively adapted IP management policies and procedures to the country's legal and economic circumstances. More than 10,000 patent grants were owned by universities in China before 2002. After passing the Act, more than 850,000 Chinese patents were owned by universities during 2002–2020 [40].

Fig. 27 shows the top ten universities from China with the highest number of Chinese patent grants during the last period 2016–2020.

PATENTS OWNED BY UNIVERSITIES

- A service creation, or work for hire, is one made by an individual when doing activities for an employer, or requiring the use of materials and technological means belonging to or supplied by the employer. In such cases, the right to apply for a patent in China for a service innovation belongs to the employing entity.

PATENTS OWNED BY THE INVENTORS

- The right to file on a patent for an invention that is not a service invention belongs to the inventor. The inventor will be the patentee until the patent application is accepted.
- For inventions where the employer and the inventor have entered into an arrangement. The terms of the signed provision shall apply to an invention made by an individual using the employer's material and technological means.

Fig. 24. Article 6 Patent law of China [38].

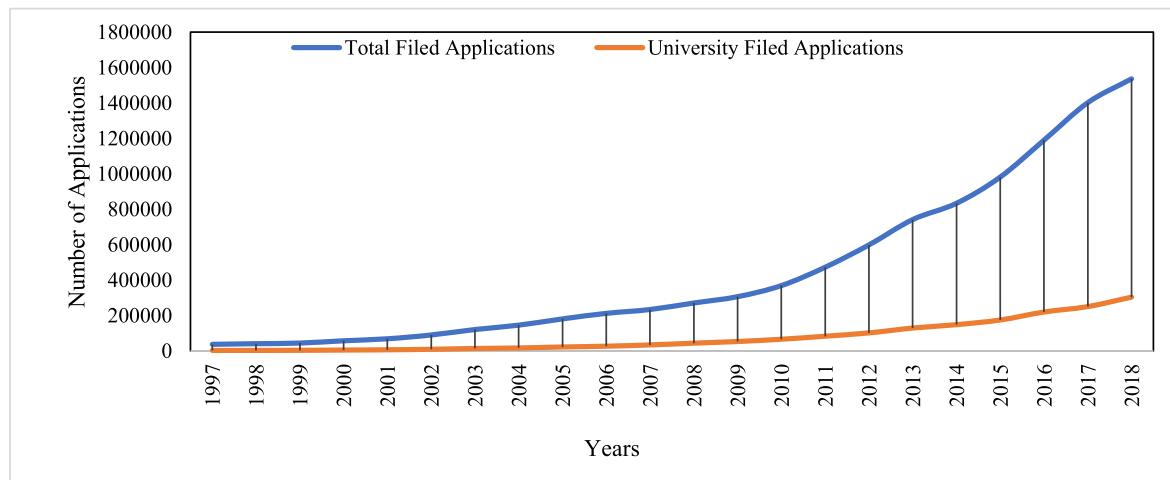


Fig. 25. Patent filings in China over the years.

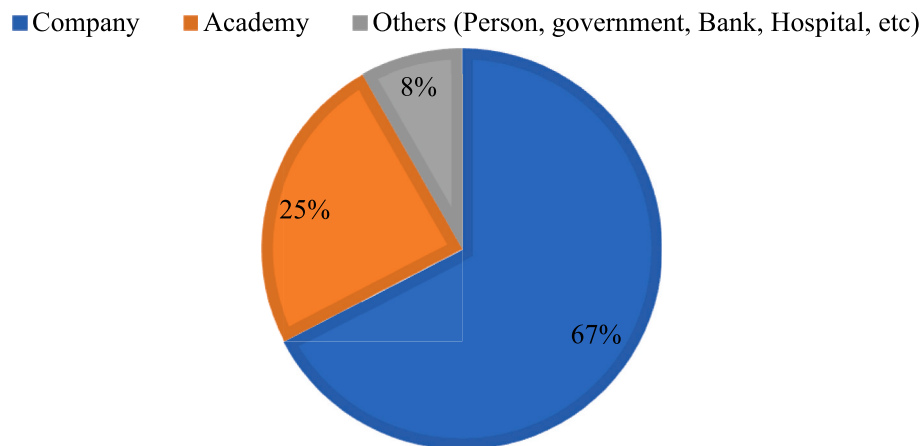


Fig. 26. Percentage distribution of Chinese patent grants from 2016 to 2020.

The data is extracted from the Patsnap database by creating a search query that focuses on the patent grant date and assignees who are solely associated with universities. Zhejiang University was found to be the first among Chinese universities, followed by Tsinghua University in second place and Southeast University in third place with highest number of Chinese patent grants. All these top Universities are China's oldest, most prestigious, and prominent higher education institutions. Zhejiang university is split into 37 campuses, schools, and divisions that

provide over 140 undergraduate and 300 graduate programs. Zhejiang University obtained more than \$680 million in research grants in 2018. More than a hundred research ventures have each earned a funding of more than US\$1.5 million [40].

Fig. 28 illustrates the top technology sectors in which Chinese universities sought patent protection. The IPC-G01 N, which belongs to control and instrumentation, was identified to have received the greatest number of Chinese patent grants for universities between 2016 and

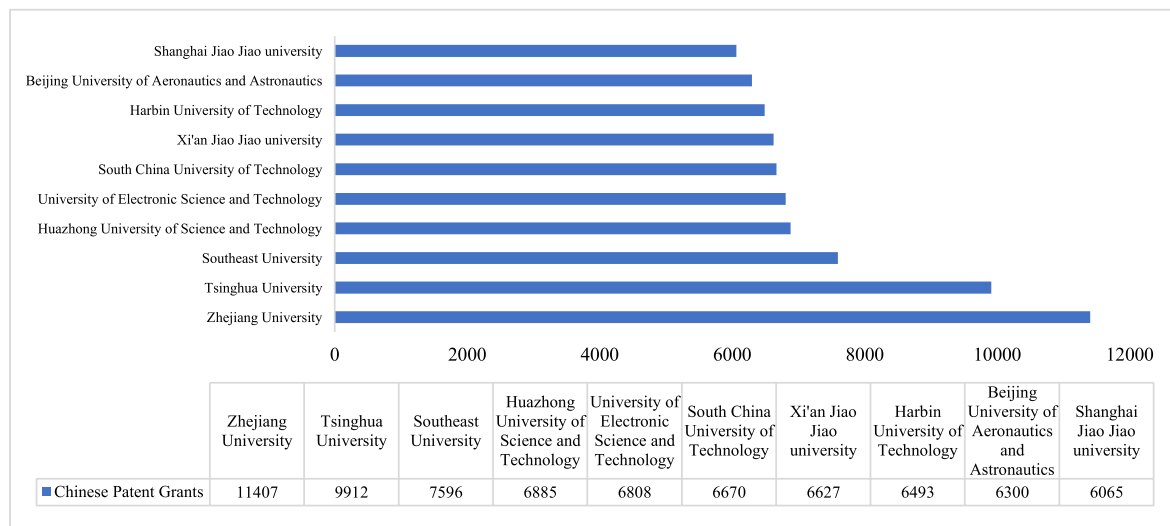


Fig. 27. Top Chinese universities in terms of patent grants from 2016 to 2020.

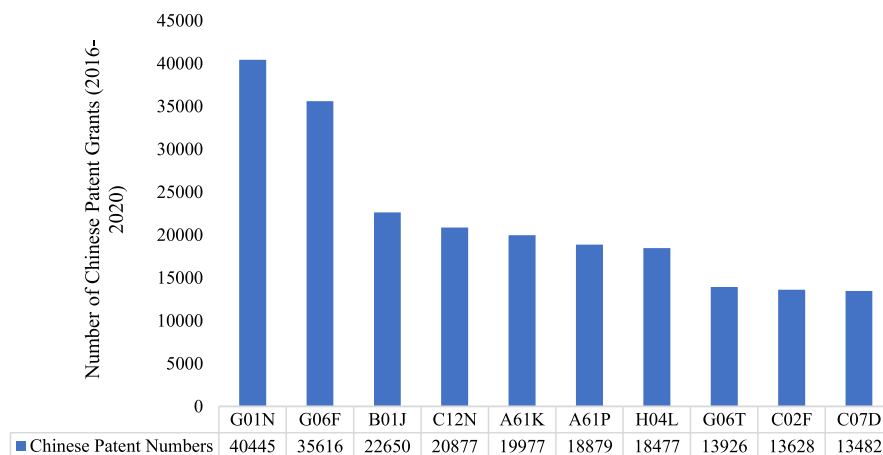


Fig. 28. Top university IPC classifications from 2016 to 2020.

Table 3

Technology Landscaping for selected 5000 granted patents.

KEY CONCEPTS DESCRIBED IN THE PATENTS	PATENTS
Power Grid, Power Supply, Distribution Network, Electric Vehicle, Energy Storage	494
Test Device, Test Piece, Optical Fiber, Light Source, Assembly	1108
Water Outlet, Water Tank, Experimental Device, Upper End, Low End	438
Alloy, Anode Material, Positive Electrode Material, Electrode Material, Sulfur Battery	778
Catalyst, Sludge, Active Component, Cement, Molecular Sieve	896
Fault Diagnosis Method, Ofdm, Vibration Signal, Input Signal, Free Control Method	820
Epoxy Resin, Graphene Oxide, Flame, Polyurethane, Copolymer	606
Second Electrode, Quantum Dot, Schottky Diode, Igbt Device, Film Transistor	373
Fluorescent Probe, Drug, Structural Formula, Synthesis, High Sensitivity	938
Gene, Amino Acid Sequence, Seq Id, Protein, Molecular Marker	623
Image, Convolutional Neural Network, Remote Sensing Image, Image Processing, Original Image	894
Network, Base Station, Node, Information, Thing	1338
Medicine, Bacteria, Traditional Chinese Medicine Composition, Culture Medium, Drug	687

2020.

Table 3 shows the technology landscaping for patents granted to different Chinese universities between 2016 and 2020. The Technology Landscape shows the layout of the technological realm, with keywords reflecting more concentrated regions of patenting activity. Technology landscaping is commonly regarded as the most efficient method of analyzing massive amounts of data pertaining to patent information. This is usually done to examine possible areas of opportunity and exploration. The patent database Patsnap is used in this research to do the technology Landscaping. This database employs text clustering, followed by keyword extraction from record text in each cluster. PatSnap will identify the most common keywords identified in that space when many comparable patents form a cluster. Table 3 shows a cluster of keywords that signify places with a high level of patent activity selected from 5000 Chinese patent grants.

2.2.1. TOP University technology transfer office in China-BASED on patents grants-Tsinghua university

Tsinghua University established Tsinghua Holdings with a registered capital of US\$395.13 million in September 2003 to consolidate industries under its supervision. Tsinghua Holdings is a Tsinghua University subsidiary. The company was founded as an internal IP asset management organization for Tsinghua's subsidiaries, which were built up by transfer of research technology to business. Tsinghua Holdings

Advanced technologies (integrated circuits, protecting the environment, and healthcare)

Innovation services

Science and technology finance

Cultural and creative industries

Online Education

Fig. 29. Key functioning areas of the TTO [41].

aspires to actively participate in and contribute to the execution of innovation-driven development plans as well as the commercialization of university scientific and technology achievements. Tsinghua Holdings serves as the university's TTO, linking it to superior industry and social resources. The TTO's major goal is to turn the university into a globally recognized scientific and technological innovation hub by providing a technological innovation investment and finance platform. Tsinghua University's scientific and technical breakthroughs have been brought to market through the TTO. They have commercialized 56 significant national technological innovations and created advances in over 62 vital technologies so far. They have helped to commercialize three key national scientific and technology initiatives and have produced outstanding achievements in technological innovation and transformation, many of which have had a worldwide influence. Fig. 29 shows the key operational areas of the TTO [41].

2.2.2. Innovation and entrepreneurship

- Tsinghua University's scientific research skills assist the TTO's commitment to the integrated development of academia, industry, and research. They have created a comprehensive research and development system that includes autonomous and continuous innovation, goods with their own IP rights, and businesses with innovation capabilities and self-owned core technologies [42].
- By the end of March 2018, the TTO has established seven national research institutions in collaboration with Tsinghua University, seven key significant research institutions, and four Ministry of Education research institutions [41].
- By the end of 2017, 56 key national breakthroughs had been commercialized, and the TTO had produced 62 technological innovations. Three major national special projects have been commercialized, and over 18,000 patents have been filed at domestic and international patent offices. Furthermore, it invested 6.95% of its revenue from royalties and startups in R&D in 2016, ranking sixth among China's top 500 organizations [41].
- Business platforms and networks of innovation cover the country to combine and strengthen innovative resources. These platforms give start-up services and inspire students and faculty to design their own businesses. By June 2017, TTO had 71 scientific and technological parks in over 70 cities and regions around the nation, 128 incubators, 7 innovation streets and 14 entrepreneurial communities. These platforms nurtured more than 6000 start-ups, over 40 of them publicly listed. They have also established business incubation facilities in Asia, North America, European Union nations and other regions [43].
- To deliver high-quality online training services and to create and innovate businesses using business resources supplied by universities, companies, and governments from across the world, the TTO created an open-ended online courses platform called the China Entrepreneurship Institute [44].

- On this online platform the students and faculty may find a variety of classes that focus on creative thinking and its related skills
- According to university policy, at least 25% of revenue earned through IP licensing must be shared with the inventor(s) in the form of cash or equity.

The TTO is comprised of 23 elite members from many sectors, including prominent entrepreneurs, investors, senior professionals, and distinguished academicians, providing advice to enterprises at various stages and trying to foster the leadership of the future of China. The mentors respond to Tsinghua industry's request for the promotion of scientific and technology innovation, business incubation and skilled leadership in business and technology [43].

3. Conclusion

The establishment of various national legislative frameworks in the USA and China, regulating university ownership of IP rights has surely benefitted academic patenting. TTOs serve an important role in motivating university faculty and students to participate in commercialization activities by providing outstanding incentives and rewards. It is critical to raise understanding of IP rights and commercialization strategies among university faculty and students to successfully accomplish the innovation cycle. Many TTOs do not have the sufficient Full-time technology transfer personnel in their offices which is critical to transform academic innovations into marketable products [45].

It takes a long time to turn an academic innovation into a commercial product. The tendency has shifted in recent decades as academics and institutions have been rewarded for their IP rights. TTOs give support and important tools to inventors who want to turn their ideas into commercial goods by cooperating with industries and raise the funds from government and private entities to advance the technologies. There are several major obstacles that inventors and TTOs confront due to the large number of stakeholders with varying interests participating in the technology transfer process. However, the overall atmosphere for academic technology transfer is improving, as different stakeholders become more skilled at aligning their interests and providing good incentives to the university inventors.

TTOs in the USA are currently efficient in terms of academic patenting and income generation per million dollars of investment. The USA universities generously compensate its inventors if their innovations are licensed, or a spin-off is created based on the patented technology. Differences in academic patenting rates might be attributed to how TTOs organize themselves internally and establish the necessary competencies to optimize commercial exploitation of inventions. Universities with strong IP policies and diverse divisions to support the technologies can have a strong influence on patenting rates and technology licensing.

The performance of a TTO is assessed by the licensing income generated by the IPs, the rate of patenting of inventions, the amount of

Table 4
Summary of TTO model assessments in various regions.

Performance Indicators	University of California	TSINGHUA University
Number of Full-time employees in the TTO	Skilled Full-time employees are available in various divisions of the TTO for the efficient technology transfer.	The TTO operates on huge funding which it receives from the Chinese government. The university website does not specify the details on how the university manage the IP and its related IP policy. The TTO staff members' details and their technical backgrounds are not listed on the university's website. CN85100113 is the first application filed by the university during Apr 01, 1985. The TTO has a long history and have successfully managed to license many breakthrough technologies from the university labs.
Age of the TTO	In 1926, the university filed its first patent application, with the application number US1926082213. The TTO is well-known and has a long history.	The TTO currently ranks second among the Chinese universities in terms of patents granted in China and PCT applications. The TTO receives huge funding from the Chinese government for patenting and they have many successful stories in licensing technologies based on university IPs
Patenting rate	The TTO currently ranks first in terms of patents granted in US and PCT filed applications.	The TTO works closely and have many collaborations with industries. It has the world's largest Science Park with many research collaboration agreements with industries.
IP Management and Licensing	The TTO has been successful in licensing out a large portfolio of IPs to companies with significant revenues. Many technology-based startups have been founded by them.	Through the PoC financing, the TTO provides very strong assistance to the projects to increase the TRL of the technologies. The TTO offers the entrepreneurship training to its faculty and students. They also provide the seed funding and incubation space for the startups.
Industrial Liaising	The TTO has a strong network with many industries.	The IP policy is not available on the university website.
Status of PoC funding to increase the TRL	Through the PoC financing, the TTO provides very strong assistance to the projects to increase the TRL of the technologies.	The TTO website is well-organized. It provides the list of technologies which are commercialized and licensed but does not display the technologies that are available for license, further I could not locate the employees in charge of handling the different divisions of the TTO.
Academic Entrepreneurship	The TTO offers the entrepreneurship training to its faculty and students. They also provide the seed funding and incubation space for the startups.	
University IP policies	The IP policy is available on the university website.	
Information on the university website	The TTO website is well-organized, and all the technologies that are offered for license are listed precisely, along with contact information for the professionals in charge. University technologies are classified according to their technological discipline. The university faculty involved in the inventions and the TTO staff details are also precisely mentioned on the website.	

sponsored research funds obtained from industry sector, and the retention of the top professors and graduate students at the university. The TTO's partnership with industry serves as the foundation for industry-sponsored research funds. It also contributes to the perception that the university promotes commercialization, which draws academics who want to make a genuine difference as well as graduate students who are more innovative. Table 4 shows the overall assessment on the TTO models based on their performance in each unit.

Declaration of interests

☑ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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